SUPPLEMENTARY MATERIAL

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Reconstructing Journal Thematic Identity from Article-Level Topics

Supplementary Material (S1–S6, tables, figures)

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Supplementary Material

Reconstructing Journal Thematic Identity from Article-Level Topics: A Reproducible Framework and a Preregistered Temporal Validation

> **Supplementary Material.** Written to the same standard as the main text. This document is the definitional and reproducibility reference for every quantity reported in the article; where a definition and the code differ, **the canonical implementation governs** and the text is corrected to match it.

S1. Formal framework: definitions and propositions

Definition 1 (methodological framework). A framework is a triple ⟨T, A, C⟩: a subfield taxonomy T (here, the OpenAlex subfield level of the topic hierarchy), an aggregation rule A that maps a set of article-level topic assignments to a distribution over T, and an applicability condition C stating the data requirements under which A may be applied to a journal (a non-empty set of assignable works within the observation window).

Definition 2 (thematic representation). For a journal *j* satisfying C over an observation window, the thematic representation is R_j = A(D_j), where D_j is the collection of the journal's article-level subfield assignments in the window. R_j is a probability distribution over T. Under the **document-weighted** mode (`p_doc`), the mass of subfield *s* is the fraction of the journal's works assigned to *s*; under the **citation-weighted** mode (`p_cit`), it is the fraction of the journal's in-window citations accruing to works assigned to *s*.

Principle 1 (observation–inference separation principle). *A representation is observed; an identity is inferred.* In full, observed representations constitute evidence, and scientific identities are inferences derived from that evidence. Applied to the object of study: any statement about a journal's thematic *identity* must be constructed from an observed representation R_j and must never replace it. Identity is reported only as an inference explicitly conditioned on ⟨T, A⟩, the window, and the data snapshot; it is never asserted as an intrinsic property and never presented as a correction of the editorial classification.

Derived objects. The **thematic nucleus** N(R_j) is defined in §S2. From R_j and the journal's editorial category set E_j ⊆ T we define the three comparative descriptors — alignment (JCA), coverage (M), and position (D) — in §S2.

Propositions

We separate propositions that are **deductive** (true by construction, given the framework) from those that are **empirical** (claims about the data, evaluated in the main text §6).

Proposition 1 (existence, deductive). Under Definitions 1–2, for every journal satisfying C the representation R_j exists and is unique given ⟨T, A, window, snapshot⟩. *Proof.* A is a well-defined function of the finite multiset D_j: each mass is a ratio of finite non-negative counts with positive denominator (guaranteed by C), and the masses sum to one by construction. Uniqueness follows from A being a function. ∎

Proposition 2 (derivability of descriptors, deductive). The nucleus and the three descriptors are computable from R_j (and, for JCA/M/D, from E_j) without any appeal to a "true" identity. *Proof.* Each is defined by an explicit deterministic operator on R_j (and E_j): the nucleus by a prefix-sum threshold (§S2), concentration and entropy by closed-form functionals, JCA by summation over E_j, coverage and position by set relations between N(R_j) and E_j. All are total functions on the domain where C holds. ∎

Proposition 3 (non-redundancy of M and D, empirical). Coverage and position are not deterministically related: knowing M does not determine D across the cohort. *Evaluated in §6.2.*

Proposition 4 (separability of JCA, empirical). Alignment is not a deterministic function of a journal's structural properties—editorial breadth (|E_j|), size (log number of works), and intrinsic shape (entropy or concentration): a non-trivial share of its cross-journal variance remains unexplained by these predictors. *Evaluated in §6.2.*

The deductive propositions guarantee that a *reproducible description* exists; every claim about what that description *reveals* is empirical and is held to the corresponding evidential standard. This boundary is the formal content of Principle 1.

S2. Metric definitions against the canonical implementation

All quantities below are produced by a single canonical implementation (§S6). The definitions state the intended object; **the executable reference is the code**, and no third-party library routine is treated as equivalent without an explicit parity check. Verified points from the code-to-documentation audit (§S5) are noted inline.

Mass distribution. For mass mode `mode ∈ {p_doc, p_cit}`, `share[s] = mass[s] / Σ_t mass[t]`, over subfields with `mass[s] > 0`. Empty or zero-total distributions are excluded by C.

Nucleus. Sort subfields by decreasing share (ties broken by ascending subfield identifier). Let `k* = min{ k : Σ_{i=1..k} share(i) ≥ τ }`, with `τ = 0.5`; the nucleus is `N = { s(1), …, s(k*) }`. The subfield whose inclusion first reaches or exceeds τ **belongs** to the nucleus. *(Audit: the code stops with `≥ τ` and includes the crossing subfield; the secondary sort key is the subfield identifier ascending.)*

Concentration. `TCI = Σ_s share[s]²` (equivalently the Herfindahl–Hirschman index of the distribution).

Entropy. `H = − Σ_s share[s] · ln share[s]`, using the **natural logarithm**; normalized entropy `H_norm = H / ln(k)` where k is the number of active subfields. *(Audit: entropy uses `ln`.)*

Jensen–Shannon divergence. For distributions p, q with mean `m = ½(p + q)`, `JSD(p, q) = ½ KL(p||m) + ½ KL(q||m)` with the Kullback–Leibler terms in **base 2** and the convention `0·log 0 = 0`; the two vectors are aligned on the union of their supports. The reported quantity is the **divergence** (bounded in [0, 1] for base-2), **not** its square root. *(Audit: `jsd()` returns the base-2 divergence, not the Jensen–Shannon distance; `scipy.spatial.distance.jensenshannon` — which returns the square-root distance — is **not** an equivalent substitute.)*

Comparative descriptors (require E_j; comparable cohort only): - **Alignment**, `JCA = Σ_{s ∈ E_j} p_cit[s]` — the citation-weighted mass of R_j falling within the editorial categories. - **Coverage**, `M` — the share of the nucleus intrinsically accounted for: (M_j = (*{s K_j E_j} p^{}*{j,s}) / (*{s K_j} p^{}*{j,s})), with nucleus (K_j) by cumulative mass (= 0.5) on (p^{}) (`spec_MD_cientifica.md`; `md_intrinsecos.py`). - **Position**, `D ∈ {nuclear, peripheral, displaced}` — with (m_E = *{s E_j} p^{}*{j,s}): **displaced** if (m_E <) ((= 0.05)); **nuclear** if (m_E) and at least 50% of (m_E) falls in (K_j) (`FRAC_CORE_NUC = 0.50`); **peripheral** if (m_E) and less than 50% of (m_E) falls in (K_j) (`spec_MD_cientifica.md`; `md_intrinsecos.py`).

Impact. `JSS` — a subfield-normalized citation score read directly from the pipeline (`discovery_cite_metrics`); not re-derived here.

Temporal stability. `S = 1 − mean_t JSD(p_doc_t, p_doc_{t+1})` over consecutive years, on the **document-weighted** distribution to avoid citation-lag bias.

Window comparison (per journal, either mode): mass divergence `JSD(R^A, R^B)`; nucleus similarity `Jaccard(N^A, N^B)`; dominant-subfield persistence `argmax R^A = argmax R^B`; and the cross-journal Pearson (and Spearman) correlations of TCI, H, and H_norm between windows. Eligibility for a comparison requires at least `MIN_SUBFIELDS = 3` active subfields in each window; the common panel is the intersection of eligible journals.

S3. Cohort construction

The cohort is built provenance-first from OpenAlex sources. Eligibility for the **sampling frame** requires a calculable subfield-normalized impact, an ISSN present in the Scopus title list, and sufficient content; this yields **1,461** journals across six subfields. The **comparable cohort** (**521**) additionally satisfies the conditions for editorial comparison (an effective editorial classification is available). The **analytic cohort** (**522**) adds one journal retained solely as an illustrative example and excluded from all aggregate estimates and from the descriptor matrix.

Two inference domains follow and are never conflated: descriptor/typology analyses (JCA, M, D) use the comparable cohort (521); intrinsic-representation analyses (concentration, stability, perturbations) use the analytic cohort (522). Reported cohort sizes are recorded per run in its manifest; no cohort size is hard-coded in the computation. *(Cross-reference: Figure 2.)*

S4. Preregistration

Two protocols were fixed, versioned, and frozen **before** the corresponding results were observed.

S4.1 Forward temporal replication (`p_cit`)

Frozen before execution: the metrics; the reference values inherited from the backward replication; the decision thresholds; and the four-cell verdict rule. A perturbation-magnitude diagnostic was added, also in advance, and its output is mandatory.

Thresholds (inherited from the backward replication A): `PISO_JSD = 0.056` , `PISO_DOM = 0.792` , `TECHO_FORMA = 0.95` , `τ = 0.5` , `MIN_SUBFIELDS = 3` , and the reference vector `REF_A = {JSD 0.046, Jaccard 1.00, dominant 0.822, TCI 0.83, H 0.90, H_norm 0.76}` .

Verdict rule (frozen four-cell grid). With `estruct_ok = (jsd_median ≤ PISO_JSD) ∧ (jaccard_median ≥ 1−ε ∧ jaccard_pct≥0.90 ≥ 0.90) ∧ (dominant_pct ≥ PISO_DOM)` and `forma_techo = min(TCI_r, H_r, Hnorm_r) > TECHO_FORMA` :

`estruct_ok`	`forma_techo`	Verdict
true	true	INCONCLUSIVE
true	false	REPRODUCES ASYMMETRY
false	true	CONTRADICTS
false	false	**WEAKENS**

No exploratory threshold tuning was performed after observing the data; the verdict follows mechanically from this grid. A verdict is registered whatever the outcome. The observed forward- `p_cit` result lands in `WEAKENS` (main text §6.4).

S4.2 Documentary control (`p_doc`)

Because the citation-calibrated grid of §S4.1 is not valid for a document-weighted metric, the documentary control uses a **separately preregistered** classification by comparison to two empirical anchors — the stable backward result and the reorganized forward- `p_cit` result — evaluated on the common panel:

```
A_CONFIRMATORY:  JSD_pdoc ≥ 0.071  AND  Jaccard≥0.90_pdoc ≤ 0.60
B_MATURATION:    JSD_pdoc ≤ 0.046  AND  Jaccard≥0.90_pdoc ≥ 0.90
INCONCLUSIVE:    any other case (intermediate region, or metrics disagree)
```

Bounds are inclusive; both metrics must indicate the same scenario, and any disagreement yields INCONCLUSIVE. The classification is a comparison against `p_cit` anchors, **not** a same-mode verdict; a formal same-mode verdict would require an anchor computed on `p_doc` . The observed documentary result is **INCONCLUSIVE** (main text §6.4, Figure 5).

S4.3 Frozen package identifiers

Artifact	SHA-256
Preregistration — forward replication (`preregistro_pruebaA_prima.md`)	`e08ce26c0a8f39b05d05aa1c1667301e79160810c5f8692f036b8862b648b982`
Canonical analysis script	`08a0e48a293096f329caed0f85ce6806b49065b4974f4f807e970747ae838769`
Manifest (`p_cit`)	`9fb560d7d53722a16125181f598cf678c2d1664ee055cacc5d39bd1322ed73ea`
Manifest (`p_doc`)	`70d1ae785a4effd759d04de984228ff989f0fb5a7cfe674914636aa197ac86b4`

Artifact	SHA-256
Runbook (`p_doc` replication; documentary-control protocol)	`36b3a95e3ec44536084aeab71bf91155cc2512a96500584db04a242e2f678363`

S5. Code-to-documentation audit

Before executing the documentary control, the canonical implementation was audited against this Supplementary Material and the frozen protocols. The audit reads functions and constants **by symbol** (not by line), records the environment and input snapshot, and classifies any reproduction failure as code / data / environment / unresolved drift. Result: **PASS**, with no drift.

Verified points (abridged): the nucleus rule (`≥ τ` , crossing subfield included; tie-break by subfield identifier ascending); the JSD (base-2 divergence, not distance); the entropy base (`ln`); the constants and reference vector; the frozen verdict grid and its four labels; eligibility and the common-panel construction; the perturbation diagnostic; the form correlations (Pearson and Spearman on the same panel); the absence of hard-coded cohort sizes in the computation; and integral reproduction of the frozen manifest at the individual-journal and aggregate levels (unexpected differences: 0; float tolerance `1e-12`). Three documentation corrections identified by the audit (nucleus defined solely by the cumulative-mass rule; explicit log bases; `ORDER BY` as optional plumbing) are incorporated in §S2.

Identity of the audited artifacts. | Field | Value | |—|—| | Canonical script SHA-256 | `08a0e48a293096f329caed0f85ce6806b49065b4974f4f807e970747ae838769` | | Canonical manifest SHA-256 | `9fb560d7d53722a16125181f598cf678c2d1664ee055cacc5d39bd1322ed73ea` | | Environment lock SHA-256 | `3e5e937b316c05aeb0637b90687e4dd98182bc3581cb1c91017dbec46770305a` | | Input snapshot identifier | `9844f2c9e1dedd7328b5299702591c7c26da6f29c39fc67bcb04b3b2e9814ff3` | | Environment | Python 3.14.5 · numpy 2.5.1 · pandas 3.0.3 · scipy 1.18.0 · PostgreSQL 16.14 |

S6. Reproducibility ledger

The analysis is released as a versioned, reproducible unit. Every reported number is traceable to the run that produced it through the identifiers below.

- **Repository:** https://github.com/Franga2026/journal-thematic-identity — canonical implementation, materialization scripts, protocols, audit runner.
- **Reserved Zenodo deposit (DOI reserved; record becomes public upon publication):** https://doi.org/10.5281/zenodo.21659764 — code, data manifests, preregistration documents, the audit report, and the SHA-256 identifier table.
- **Version identifiers:** the frozen manuscript, the preregistrations, the canonical script, the manifests, and the audit each carry a SHA-256 hash (tables in §S4.3 and §S5); the manuscript version register (`VERSION.md`) records them and the outcome of each execution.

Reproduction procedure: instantiate the environment (§S5), point the pipeline at the frozen input snapshot, and re-run the canonical script; the regression gate reproduces the frozen manifest before any new analysis is permitted.

S-Figures / S-Tables

Table S1. Subfield composition of the cohort (sampling frame = `in_content` on Path-1 six; source: `journal_frame_index`).

Subfield code	Sampling frame (`in_content`)	Comparable (`frame_status='comparable'`)
1311	83	49
2604	130	53
2716	6	1
3106	50	25
3109	28	8
3312	1164	385
Total	**1461**	**521**

Table S2. Reproduction diff from the code-to-documentation audit (`audit_A_prima_consistency/audit_consistency_report.json`, control `10_manifiesto`): unexpected aggregate differences = **0**; unexpected individual (per-journal) differences = **0**; float tolerance = `1e-12`.

- **Figure S1.** Continuous vs threshold-based observables — a schematic of how a nucleus set can turn over under a mass change too small to move the divergence (the operator-decoupling made explicit for §7).
- **Figure S2.** Perturbation-magnitude diagnostic (distribution of per-journal divergences under the forward replication).

Supplementary Material · definitional and reproducibility reference for the manuscript.